

HARSH KUMAR KARN

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PROFESSIONAL SUMMARY

2026 B.Tech IT graduate with a track record across full-stack development, backend engineering, AI/agentic systems, and data analysis. Comfortable owning a feature from database schema through API to a deployed React/Next.js frontend, building RAG and agentic pipelines on top of LLM APIs (OpenAI, Claude, Gemini), and turning raw data into SQL-backed reporting and dashboards stakeholders actually use. Picks up a new stack quickly and enjoys contributing wherever a project needs it.

EDUCATION

Rungta College Of Engineering And Technology

B.Tech - Information Technology

2022 – 2026

Bhilai, Chhattisgarh

TECHNICAL SKILLS

- **Full-Stack & Languages:** Python, JavaScript/TypeScript, SQL, React.js, Next.js, Tailwind CSS
- **Backend & APIs:** FastAPI, Node.js/Express, REST API design, JWT auth, WebSockets/Socket.IO, microservices
- **AI & Agentic Systems:** LangChain, RAG (FAISS/BM25), prompt engineering, agentic workflows, OpenAI/Claude/Gemini/Groq APIs
- **Data & Analytics:** Pandas, NumPy, EDA, SQL (CTEs, window functions), Power BI, Tableau, Excel/VBA
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, Redis, AWS, Docker, GitHub Actions CI/CD

EXPERIENCE

Bhilai Steel Plant

Data Analyst Intern

09 2025 – 10 2025

Bhilai, Chhattisgarh

- Built Python/Pandas ETL pipelines processing 10,000+ records, cutting processing time by 35%; authored complex SQL (CTEs, window functions) resolving 200+ data integrity issues.
- Delivered Power BI/Tableau dashboards for senior stakeholders; tracked tasks in JIRA, maintained Confluence docs in Agile sprints with fast-changing requirements.

PROJECTS

Fynlo - UPI Expense Platform | FastAPI, React/Next.js, PostgreSQL, Celery, Redis, Docker

- Built a full-stack app end to end: a **FastAPI backend** with JWT-authenticated REST APIs and a PostgreSQL/SQLAlchemy data layer, paired with a **React/Next.js frontend** for dashboards and file uploads.
- Designed an async pipeline (Celery + Redis) that batches bank-statement transactions through the Gemini API for categorization without blocking API responses; Dockerized the full stack. [Repo](#)

SHORTR - URL Shortener SaaS | FastAPI, PostgreSQL, Redis, ClickHouse, Docker

- Built a **backend-heavy SaaS** in FastAPI: PostgreSQL (SQLAlchemy + Alembic) for link ownership and Redis edge caching for sub-millisecond redirects at high traffic.
- Logged click events into ClickHouse through an async pipeline separate from the redirect path, and implemented dual auth (JWT + API keys) for dashboard and programmatic access. [Repo](#)

PulseRAG - QuantGPT Engine | FastAPI, FAISS, BM25, Sentence-Transformers

- Built a production **RAG pipeline** for financial documents: hybrid dense FAISS + sparse BM25 retrieval with freshness-decay re-ranking, averaging under 200ms per query.
- Split the system into independently replaceable retriever, ranker, and generator services, so embedding or LLM providers can be swapped without breaking the API contract. [Repo](#)

Airlines Data Analysis | Python, SQL (SQLite), Pandas, Streamlit

- Modeled a 6-table relational schema and wrote SQL (CTEs, JOINS, window functions) to analyze revenue and profit drivers across multiple data sources.
- Identified a 10% occupancy uplift opportunity through statistical analysis and built an interactive Streamlit dashboard for automated reporting. [Live site](#)

CERTIFICATIONS & ACHIEVEMENTS

- Data Analytics - Coding Spoon (Python, Data Cleaning, EDA, Visualization)
- API Prototyping Learning Path – Postman Academy (API Design, Testing & Documentation)
- The Complete Web Development Bootcamp - Udemy (Full-Stack: HTML, CSS, JS, Node.js, React, MongoDB)
- Anthropic Education - AI Fluency: Framework & Foundations, Claude 101, Claude Code in Action